determine the possible mass and force charges, they require a total of 10 spacetime dimensions, their curled-up dimensions must be in one of the Calabi-Yau shapes, etc.—and for this reason we have not emphasized their differences in previous chapters. Nevertheless, analyses in the 1980s showed that they do differ. You can read more about their properties in the endnotes, but it's enough to know that they differ in how they incorporate supersymmetry as well as in significant details of the vibrational patterns they support.2 (Type I string theory, for example, has open strings with two loose ends in addition to the closed loops we have focused on.) This has been an embarrassment for string theorists because although it's impressive to have a serious proposal for the final unified theory, having five proposals takes significant wind from the sails of each.

The second deviation from inevitability is more subtle. To fully appreciate it, you must recognize that all physical theories consist of two parts. The first part is the collection of fundamental ideas of the theory, which are usually expressed by mathematical equations. The second part of a theory comprises the solutions to its equations. Generally speaking, some equations have one and only one solution while others have more than one solution (possibly many more). (For a simple example, the equation "2 times a particular number equals 10″ has one solution: 5. But the equation "0 times a particular number equals 0″ has infinitely many solutions, since 0 times any number is 0.) And so, even if research leads to a unique theory with unique equations, it might be that inevitability is compromised because the equations have many different possible solutions. By the late 1980s, it appeared that this was the case with string theory. When physicists studied the equations of any one of the five string theories, they found that they do have many solutions—for example, many different possible ways to curl up the extra dimensions—with each solution corresponding to a universe with different properties. Most of these universes, although emerging as valid solutions to the equations of string theory, appear to be irrelevant to the world as we know it.

These deviations from inevitability might seem to be unfortunate fundamental characteristics of string theory. But research since the mid-1990s has given us dramatic new hope that these features may be merely reflections of the way string theorists have been analyzing the